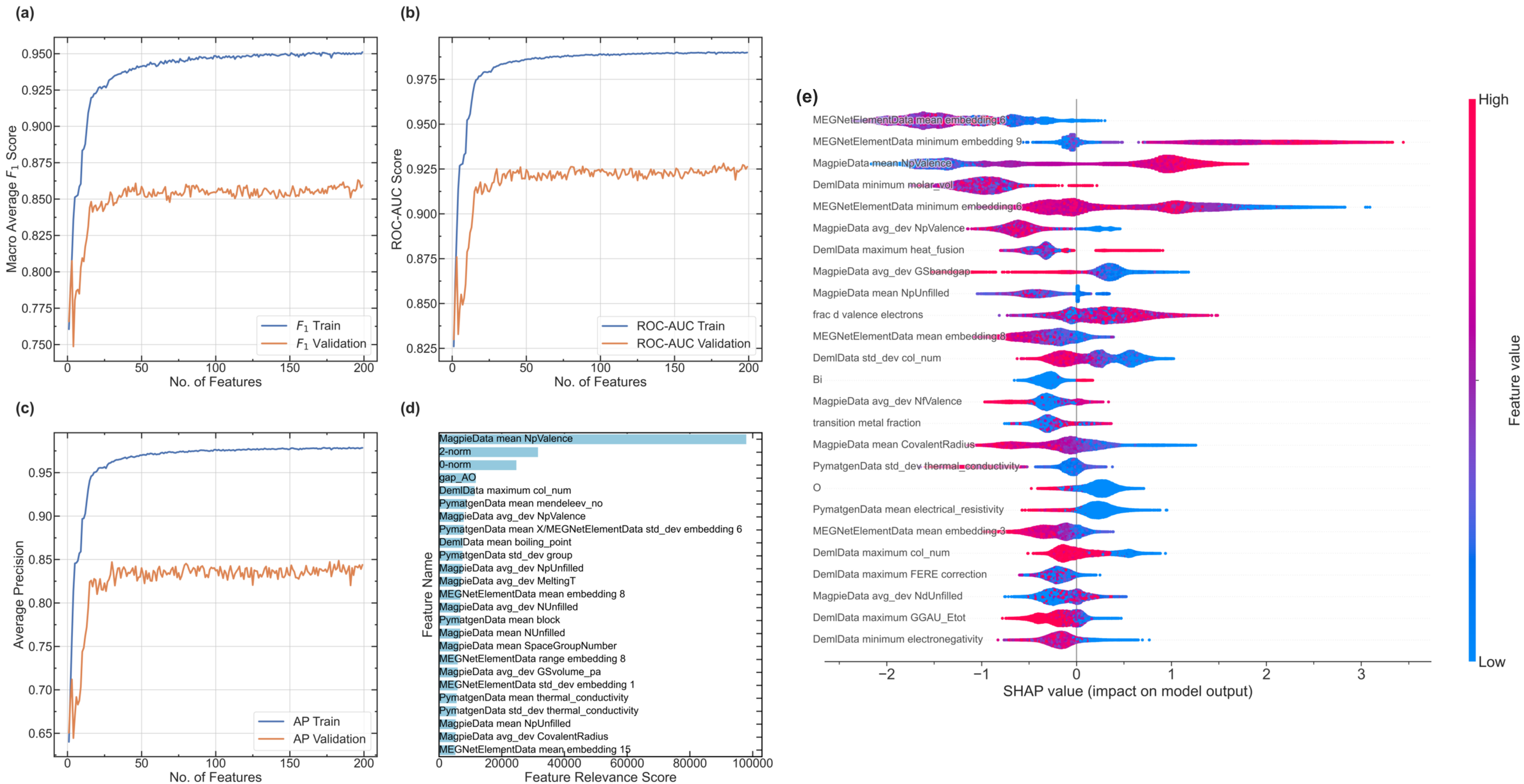
(a)
Macro Average F_1 Score
No. of Features
F_1 Train
F_1 Validation
(b)
ROC-AUC Score
No. of Features
ROC-AUC Train
ROC-AUC Validation
(c)
Average Precision
No. of Features
AP Train
AP Validation
(d)
Feature Name
Feature Relevance Score
MagpieData mean NpValence
2-norm
0-norm
gap_AO
DemlData maximum col_num
PymatgenData mean mendeleev_no
MagpieData avg_dev NpValence
PymatgenData mean X/MEGNetElementData std_dev embedding 6
DemlData mean boiling_point
PymatgenData std_dev group
MagpieData avg_dev NpUnfilled
MagpieData avg_dev MeltingT
MEGNetElementData mean embedding 8
MagpieData avg_dev NUnfilled
PymatgenData mean block
MagpieData mean NUnfilled
MagpieData mean SpaceGroupNumber
MEGNetElementData range embedding 8
MagpieData avg_dev GSvolume_pa
MEGNetElementData std_dev embedding 1
PymatgenData mean thermal_conductivity
PymatgenData std_dev thermal_conductivity
MagpieData mean NpUnfilled
MagpieData avg_dev CovalentRadius
MEGNetElementData mean embedding 15
(e)
MEGNetElementData mean embedding 6
MEGNetElementData minimum embedding 9
MagpieData mean NpValence
DemlData minimum molar_vol
MEGNetElementData minimum embedding 6
MagpieData avg_dev NpValence
DemlData maximum heat_fusion
MagpieData avg_dev GSbandgap
MagpieData mean NpUnfilled
frac d valence electrons
MEGNetElementData mean embedding 8
DemlData std_dev col_num
Bi
MagpieData avg_dev NfValence
transition metal fraction
MagpieData mean CovalentRadius
PymatgenData std_dev thermal_conductivity
O
PymatgenData mean electrical_resistivity
MEGNetElementData mean embedding 3
DemlData maximum col_num
DemlData maximum FERE correction
MagpieData avg_dev NdUnfilled
DemlData maximum GGAU_Etot
DemlData minimum electronegativity
SHAP value (impact on model output)
Feature value
High
Low